

# Samuel Schiavone

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## PROFILE

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I am a researcher, programmer, and mathematician who combines mathematical abstraction with computational tools in order to discover, analyze, and elucidate. I have experience in learning abstract results and applying them to produce practical solutions.

## SKILLS

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**Programming Languages:** Python, PostgreSQL, SageMath, Magma, R

**Programming Tools:** Git, Flask, Jinja, GNU Parallel, Scikit-learn, Pandas

**Languages:** English (native), French (fluent), Spanish (intermediate)

## SELECTED PROJECTS

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### Development for the L-functions and modular forms database (LMFDB)

May 2018 – Present

*Full Stack Web Developer*

*Remote*

- Collaborated on a joint open-source project to create an accessible repository of data in algebra and number theory, used by over 30,000 researchers and educators in math
- Co-created two sections of an online database: added data to the backend using Python and PostgreSQL; designed webpages to display the data using Flask, Jinja, and HTML
- Supervised four undergraduate students in the creation of a Magma and SageMath package to compute additional data
- Connected to other sections of the LMFDB by identifying and linking to related objects using scripts written in Python and SageMath

### Computations in Inverse Galois Theory

Jan 2022 – Nov 2024

*Academic Researcher*

*Cambridge, Massachusetts*

- Collaborated on a team of 6 researchers to solve an open mathematical problem in computational number theory
- Combined theoretical expertise with practical programming skills in Python, PostgreSQL, SageMath, and Magma to produce a research article and two repositories of code, one a general Magma package and one tailored to a specific example
- Presented results in two invited talks to audiences of 30-50 researchers at VaNTAGe and Simons AGNTC annual meeting
- Synthesized detailed understanding of theoretical results in arithmetic geometry with technical programming expertise to give an explicit solution to an open mathematical problem

## EXPERIENCE

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### Research Scientist

Sep 2019 – Aug 2025

*Massachusetts Institute of Technology*

*Cambridge, Massachusetts*

- Conducted research in computational arithmetic geometry
- Devised and implemented algorithms to compute mathematical objects, wrote mathematical proofs, and chronicled my research in academic articles
- Aided in the development and maintenance of the L-functions and Modular Forms Database (<https://www.lmfdb.org/>), working both with data in PostgreSQL and Python on the backend and in Flask, Jinja, and HTML on the frontend
- Instructed undergraduate students in courses on linear algebra, algebraic geometry, mathematical communication, and Belyi maps

## EDUCATION

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### Dartmouth College

*Ph.D. in Mathematics*

### The University of Vermont

*M.S. in Mathematics*

### Amherst College

*B.A. in Mathematics, magna cum laude*

Hanover, New Hampshire

*Sep 2019*

Burlington, Vermont

*Dec 2013*

Amherst, Massachusetts

*May 2010*